EMPLOYEE SALARY ANALYSIS

1. Introduction

Employee Salary Analysis is a beginner-level Python project that summarizes employee salary data. It calculates the total, average, highest, and lowest salary from a manually created dataset.

2. Objective

The objective is to practise numerical data processing in Python, use built-in functions for basic calculations, and compare computed statistics with the original data.

3. Problem Statement

Calculating salary statistics manually can be repetitive and error-prone. This program automates the calculations and presents the results in a clear, repeatable format.

4. Dataset

The suggested dataset is `employee_salaries.csv`, containing the columns `Employee_ID`, `Employee_Name`, and `Salary`.

The example values are synthetic and created for demonstration; they do not represent real employees.

5. Methodology

The project follows these steps:

1. Read employee salary records from a CSV file.
2. Validate that salary values are numeric and non-negative.
3. Calculate the employee count, total salary, average salary, highest salary, and lowest salary.
4. Identify the employees associated with the highest and lowest salaries.
5. Display the results and compare them with the source records.

6. Calculations

Total Salary: The sum of all valid salary values.

Average Salary: Total salary divided by the number of employees.

Highest Salary: The maximum salary value in the dataset.

Lowest Salary: The minimum salary value in the dataset.

7. Testing Plan

Test the program with ordinary salary values, a single employee, repeated salary values, an empty dataset, non-numeric salary text, and negative salary values.

The program should display correct statistics for valid records and clearly report invalid data.

Actual test results should be recorded after executing the tests.

8. Expected Outcome

The program should report the number of employees and the total, average, highest, and lowest salary.

It should also identify the employee or employees associated with the maximum and minimum salary values.

9. Concepts Demonstrated

The project demonstrates Python functions, CSV processing, loops, exception handling, numerical calculations, input validation, and structured output.

Pandas may be used as an optional extension.

10. Limitations and Future Improvements

The demonstration dataset is small and synthetic.

Future versions could include department-wise salary analysis, salary distribution charts, comparisons across departments, and exporting summary results to a separate CSV file.

11. Conclusion

This project demonstrates how Python can automate basic salary analysis and reduce manual calculation effort.

Before submission, run the code, record actual test outcomes, and add the real GitHub and LinkedIn URLs to the repository documentation.

12. Project Links

GitHub Repository: https://github.com/venky1318/employee-salary-analysis